Recitation 3

Gradient Descent, Gaussian Naive Bayes, Logistic Regression

10-701: Introduction to Machine Learning

Feb 6, 2026

1 Gradient Descent

Gradient descent (GD) is one of the most commonly used optimization algorithms in machine learning. Here we will go over 1) why we are using gradients in the first place and 2) why stochastic gradient descent (SGD) works.

1.1 Gradient Points in the Direction of Steepest Ascent

In class, we saw a pictorial sketch of why gradient descent makes sense; we will discuss it more formally here.

One way of thinking about this is that for a differentiable multivariate function $f : \mathbb{R}^d \to \mathbb{R}$, the gradient at point $\boldsymbol{x} \in \mathbb{R}^d$ (i.e. $\nabla f(\boldsymbol{x})$) points in the direction of *steepest ascent* at $\boldsymbol{x}$.

Recall from calculus that we define the directional derivative with respect to some unit vector $\boldsymbol{u} \in \mathbb{R}^d$ as

$$D_{\boldsymbol{u}} f(\boldsymbol{x}) = \lim_{h \to 0} \frac{f(\boldsymbol{x} + h\boldsymbol{u}) - f(\boldsymbol{x})}{h}$$

In words, the directional derivative describes how the function value instantaneously changes if we step along the direction of $\boldsymbol{u}$ from point $\boldsymbol{x}$. With some calculation, one can show that

$$\begin{aligned} D_{\boldsymbol{u}} f(\boldsymbol{x}) &= \boldsymbol{u}^T \nabla f(\boldsymbol{x}) \\ &= ||\boldsymbol{u}|| ||\nabla f(\boldsymbol{x})|| \cos\theta \ (\because \text{inner product}) \end{aligned}$$

where θ is the angle between $\boldsymbol{u}$ and $\nabla f(\boldsymbol{x})$. We know $||\boldsymbol{u}|| = 1$, and it is easy to see that when $\theta = 0$, $D_{\boldsymbol{u}} f(\boldsymbol{x})$ is maximized.

Thus, at each point, we should step in the direction of the gradient to maximally *increase* the function value (gradient ascent). Meanwhile, we should step in the direction opposite of the gradient to maximally *decrease* the function value (gradient descent). Whether we want to use gradient ascent or descent will depend on whether we want to maximize or minimize some objective function.

1.2 Stochastic Gradient Descent

In lecture, you are introduced with the concept of gradient descent (GD). However, the *stochastic* gradient descent (SGD) is more common in practice than the vanilla GD, as the

gradient updates in SGD are computationally cheaper when working with large datasets but still lead to good, generalizable solutions (often even better). An in-depth discussion of SGD is beyond the scope of this course, but here we will see why it makes sense at a high level.

Suppose we have some ML model (e.g., linear regression, logistic regression, neural network) and we want to optimize the parameters $\boldsymbol{w}$ of that model using data $\mathcal{D} = \{(\boldsymbol{x}_i, \boldsymbol{y}_i)\}_{i=1}^n$. And let's say that our loss function is

$$\mathcal{L}_{\text{total}} = \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}(\boldsymbol{x}_i, \boldsymbol{y}_i)$$

which is an average of the loss $\mathcal{L}$ for each data point $(\boldsymbol{x}_i, \boldsymbol{y}_i)$ across our dataset.

Now, we want to use the following SGD algorithm to iteratively update $\boldsymbol{w}_t$:

1. Randomly sample (without replacement) m indices from $\{1, \ldots, n\}$. Call the set of sampled indices $\mathcal{B}$.
2. Calculate the loss using the sampled data points

   $\tilde{\mathcal{L}} = \frac{1}{m} \sum_{i=1}^{n} \mathcal{L}(\boldsymbol{x}_i, \boldsymbol{y}_i) \underbrace{1[i \in \mathcal{B}]}_{\text{indicator}}$
3. Update $\boldsymbol{w}_{t+1} \leftarrow \boldsymbol{w}_t - \eta \frac{\partial \tilde{\mathcal{L}}}{\partial \boldsymbol{w}}$

Based on this setup, answer the question below.

1. In statistics, the bias of an estimator (or bias function) is the difference between this estimator's expected value and the true value of the parameter being estimated.

   $$Bias(\hat{\theta}, \theta) = \mathbb{E}_{x|\theta}[\hat{\theta} - \theta]$$

   An estimator or decision rule with zero bias is called unbiased. Show that the stochastic gradient is an unbiased estimator of the gradient, i.e. show that:

   $$\mathbb{E}\left[\frac{\partial \tilde{\mathcal{L}}}{\partial \boldsymbol{w}}\right] = \frac{\partial \mathcal{L}}{\partial \boldsymbol{w}}$$

   (Hint: $\mathbb{E}\big[1[i \in \mathcal{B}]\big] = p(i \in \mathcal{B})$. Also check out the remark below if you need better intuition.)

Remark: Randomly choosing the data points for updating the parameters at each iteration is what makes SGD *stochastic*. Try comparing $\frac{\partial \mathcal{L}}{\partial \boldsymbol{w}}$ and $\frac{\partial \tilde{\mathcal{L}}}{\partial \boldsymbol{w}}$; you should notice that since we are only using a random subset of all of our training points to calculate the gradient, $\frac{\partial \tilde{\mathcal{L}}}{\partial \boldsymbol{w}}$ can be thought of as an approximation of $\frac{\partial \mathcal{L}}{\partial \boldsymbol{w}}$.

2 Naive Bayes

2.1 Sample Problem

Suppose that there are d binary features $(X_1, X_2, \ldots, X_d)$. Assume d is even. Consider the following pairing of the d features:

$$(X_1, X_2), (X_3, X_4), \ldots, (X_{d-1}, X_d)$$

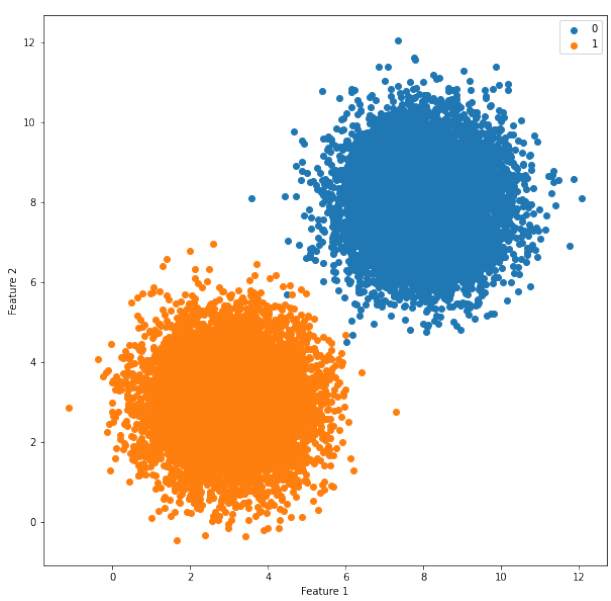
For each of the above pairs (X_i, X_{i+1}), assume X_i and X_{i+1} are **dependent**. However, assume the 2 pairs (X_i, X_{i+1}) and (X_j, X_{j+1}) are themselves **independent** when $i \neq j$ given the class.

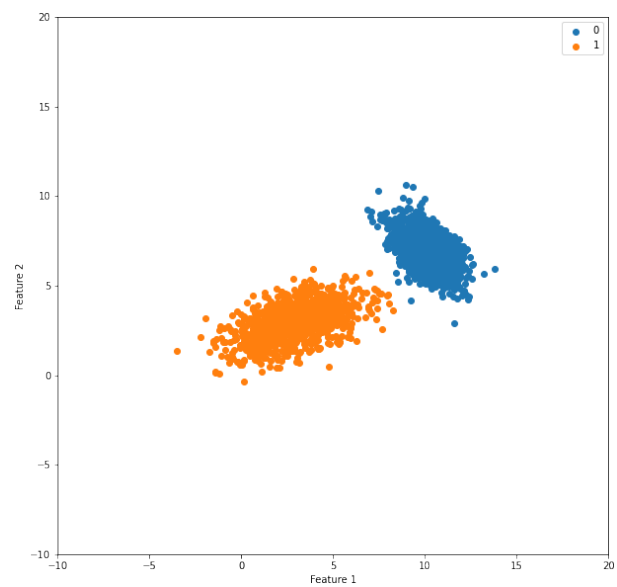
If the class-conditional distribution of each pair of features is known (as well as the class prior), is it possible to construct a classification algorithm using the Naive Bayes approach? If it's possible, how would you do it? If it's not possible, why is it not?

2.2 Gaussian Contour Plots

For a one-dimensional Gaussian, the probability density looks similar to bell curve. For a two-dimensional Gaussian, if both coordinates are independent of one another then the density concentrates in circles. If the two coordinates are not independent, then the density will look elliptical like in the figure above.

For each dataset below, determine if the Naive Bayes assumption is valid. Assume that the data given the class label is distributed as a multivariate Gaussian.





2.3 Exam Style Practice Problems

1. **Select All that Apply:** In a Naive Bayes problem, suppose we are trying to compute $P(Y|X_1, X_2, X_3, X_4)$. Furthermore, suppose X_2 and X_3 are identical (i.e., X_3 is just a

copy of X_2). Finally, assume X_2 is not independent of Y. Which of the following are true in this case?

- ☐ Naive Bayes will learn identical parameter values for $P(X_2|Y)$ and $P(X_3|Y)$.
- ☐ Naive Bayes will predict $P(Y|X_1, X_2, X_3, X_4) < P(Y|X_1, X_2, X_4)$.
- ☐ Naive Bayes will predict $P(Y|X_1, X_2, X_3, X_4) > P(Y|X_1, X_2, X_4)$.
- ☐ None of the above

2. **Select All that Apply:** Gaussian Naive Bayes, in general, can learn non-linear decision boundaries. Consider the simple case where we have just one real-valued feature $X_1 \in \mathbb{R}$ from which we wish to infer the value of label $Y \in \{0, 1\}$. The corresponding generative story would be:

   $Y \sim \text{Bernoulli}(\phi)$
   $X_1 \sim \text{Gaussian}(\mu_y, \sigma_y^2)$
   where the parameters are the Bernoulli parameter ϕ and the class-conditional Gaussian parameters $\mu_0, \sigma_0^2, \mu_1, \sigma_1^2$ corresponding to Y = 0 and Y = 1 , respectively.

   Consider a linear decision boundary in one dimension described by the rule: if $X_1 > c$, then $Y = 1$, else $Y = 0$, where c is a real-valued threshold. Is it possible (in the 1D case) to construct a Gaussian Naive Bayes classifier with a decision boundary that cannot be expressed by a rule in the above form?

   - ☐ Yes, this can occur if the Gaussians are of equal means and equal variances.
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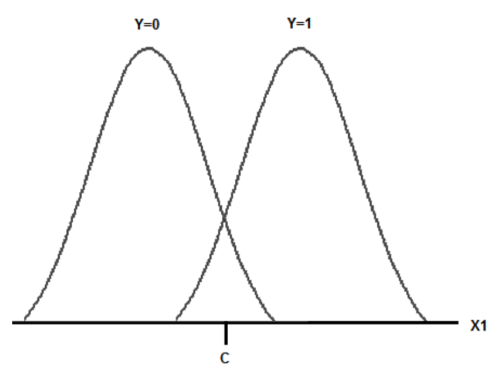
   - ☐ Yes, this can occur if the Gaussians are of unequal means and unequal variances.
   - ☐ No, it is not possible.



If X1>C then Y=1, else Y=0

3 Logistic Regression

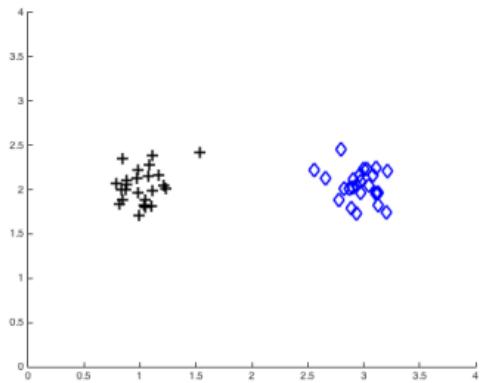
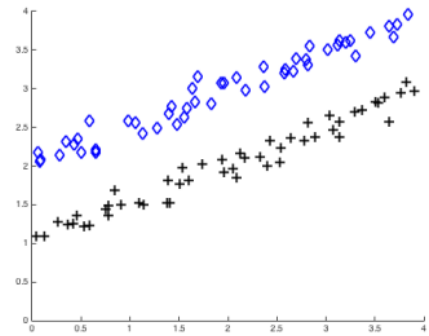
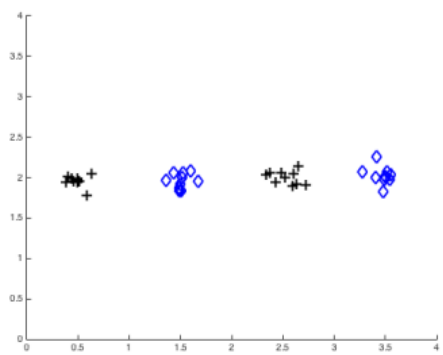
1. For a probability value $p \in (0, 1)$, what is the range of the odds, $\frac{p}{1-p}$, and the log-odds, $\log\left(\frac{p}{1-p}\right)$? Explain why this makes the log-odds a desirable transformation of our data to fit with our affine model.

2. We consider the following models of logistic regression for a binary classification with a sigmoid function $g(z) = \frac{1}{1+e^{-z}}$:

   - Model 1: $P(Y = 1 \mid X, w_1, w_2) = g(w_1 X_1 + w_2 X_2)$
   - Model 2: $P(Y = 1 \mid X, w_0, w_1, w_2) = g(w_0 + w_1 X_1 + w_2 X_2)$

   We have three training examples:

   $$\begin{array}{lll} x^{(1)} = [1, 1]^T & x^{(2)} = [1, 0]^T & x^{(3)} = [0, 0]^T \\ y^{(1)} = 1 & y^{(2)} = 0 & y^{(3)} = 1 \end{array}$$

   Does it matter how the third example is labeled in Model 1? i.e., would the learned value of $w = (w1, w2)$ be different if we change the label of the third example to 0? Does it matter in Model 2? Briefly explain your answer.

3. For each of the following figures, compare the performance of a Logistic Regression model vs a Gaussian Naive Bayes model. To be more precise, if a model is able to classify the given points, indicate the appropriate decision boundary. If not, provide a reasonable justification as to why the model is unable to classify the particular set of points.